

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 November 2023
2 hours 30 minutes

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Evaluate $7 + 6 \times 3 - 6$. [3]
2. Evaluate $-5 - (-14)$. [3]
3. Work out $1 + \frac{1}{5}$. Give your answer as a mixed number if possible. [3]
4. Find 25% of 200. [3]
5. Solve $3x + 5 = 17$. [3]
6. Share \$36 in the ratio 2 : 7. [4]
7. Find (a) the HCF and (b) the LCM of 18 and 28. [4]
8. Write 9000000 in standard form. [3]
9. Simplify $5^2 \times 5^2$. Leave your answer as a power of 5. [3]
10. $n(A) = 11$, $n(B) = 10$ and $n(A \cap B) = 4$. Find $n(A \cup B)$. [3]
11. Expand and simplify $2(x + 1) - 6x$. [3]
12. Factorise completely $18x + 24$. [3]
13. Factorise $x^2 + 10x + 24$. [4]
14. Make h the subject of the formula $A = 2\pi r(r + h)$. Take π as a symbol. [4]
15. Solve $2x + 2 < 12$. [3]
16. Solve the simultaneous equations
$$\begin{aligned}x + y &= 7 \\x - y &= 5.\end{aligned}$$
[4]
17. The n th term of a sequence is $2 + 4(n - 1)$. Find the 8th term. [3]
18. In triangle ABC, angle A = 57° and angle B = 69° . Find angle C. [3]
19. A regular polygon has 9 sides. Find the size of each exterior angle. [3]
20. A right-angled triangle has shorter sides 9 cm and 12 cm. Calculate the hypotenuse. [4]
21. In an isosceles right-angled triangle, the two equal sides are 6 cm. Write $\tan 45^\circ$ and hence find the opposite side to 45° if the adjacent is 6 cm. [4]

- 22.** A triangle has base 8 cm and perpendicular height 5 cm. Find its area. **[3]**
- 23.** Calculate the area of a circle of radius 21 cm. Take $\pi = 22/7$. **[4]**
- 24.** A cuboid measures 6 cm by 7 cm by 2 cm. Find its volume. **[3]**
- 25.** Find the gradient of the line joining (1, 1) and (3, 7). **[3]**
- 26.** Find the midpoint of the line joining (1, 1) and (3, 7). **[3]**
- 27.** A bag contains 7 red, 6 blue and 5 green beads. A bead is drawn at random. Find the probability that it is red. **[3]**
- 28.** Find the mean of 4, 11, 6, 11, 4. **[3]**
- 29.** Work out $\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ where $\begin{pmatrix} \\ \end{pmatrix}$ denotes a 2×2 matrix times a column vector. **[4]**
- 30.** A trader in Mbare buys a crate of tomatoes for \$25 and sells it for \$30. Calculate the percentage profit. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).